

Revisiting DFSMShsm Hints and Tips

Edward Baker

March 8th, 2006, Session 3031

ebaker@us.ibm.com

Legal Disclaimer



Copyright © 2006 by International Business Machines Corporation.

No part of this document may be reproduced or transmitted in any form without written permission from IBM Corporation.

Product data has been reviewed for accuracy as of the date of initial publication. Product data is subject to change without notice. This information could include technical inaccuracies or typographical errors. IBM may make improvements and/or changes in the product(s) and/or programs(s) at any time without notice.

Any statements regarding IBM's future direction and intent are subject to change or withdrawal without notice, and represent goals and objectives only.

References in this document to IBM products, programs, or services does not imply that IBM intends to make such products, programs or services available in all countries in which IBM operates or does business. Any reference to an IBM Program Product in this document is not intended to state or imply that only that program product may be used. Any functionally equivalent program, that does not infringe IBM's intellectually property rights, may be used instead. It is the user's responsibility to evaluate and verify the operation of any non-IBM product, program or service.

THE INFORMATION PROVIDED IN THIS DOCUMENT IS DISTRIBUTED "AS IS" WITHOUT ANY WARRANTY, EITHER EXPRESS OR IMPLIED. IBM EXPRESSLY DISCLAIMS ANY WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE OR NONINFRINGEMENT. IBM shall have no responsibility to update this information. IBM products are warranted according to the terms and conditions of the agreements (e.g., IBM Customer Agreement, Statement of Limited Warranty, International Program License Agreement, etc.) under which they are provided. IBM is not responsible for the performance or interoperability of any non-IBM products discussed herein.

The provision of the information contained herein is not intended to, and does not, grant any right or license under any IBM patents or copyrights. Inquiries regarding patent or copyright licenses should be made, in writing, to:

IBM Director of Licensing
IBM Corporation
North Castle Drive
Armonk, NY 10504-1785
U.S.A.

Trademarks



The following are trademarks of the International Business Machines Corporation in the United States and/or other countries.

DFDSS

DFHSM

DFSMSdfp

DFSMSHsm

DFSMSdss

IBM®

System/390®

SNAP/SHOT®

Enterprise Storage Server®

FlashCopy®

OS/390®

z/OS®

Agenda



- Managing and Tuning the **Recall** Environment
 - ▶ ARCRPEXT
 - ▶ Common Recall Queue
 - ▶ Reducing Recall Contention
- Managing and Tuning the **Migration** Environment
 - ▶ Fast Subsequent Migration
- Managing and Tuning the **Backup** Environment
 - ▶ Backup Direct to Tape
 - ▶ Increased Backup Tasks
- Managing and Tuning the **HSM Control Data Sets** Environment
 - ▶ CDS Care and Feeding
 - ▶ Backing up your Control Data Sets
 - ▶ CDS Recovery Planning

Agenda



■ Managing and Tuning the **HSM Tape** Environment

- ▶ Full Volume Dump Stacking
- ▶ ABARS File Stacking
- ▶ Collocate ABARS ML2 data
- ▶ Reduce Tape Spanning
- ▶ Recycle TAPELIST and SELECT
- ▶ Recycle Performance

■ Misc Tips and Hints

- ▶ MASH
- ▶ SMF Consolidation
- ▶ Avoid Log Overhead
- ▶ Use of the ONLYIF command in PARMLIB member

Prioritizing Recalls

- ARCRPEXT can be used to prioritize recalls, deletes and recovers of data sets
- Priority range 0-100
- Wait Type ahead of NoWait Type
- Recalls and Deletes on the same queue

Wait
Type

RECALL
Prod123

80

RECALL
TSO888

50

RECALL
Dev456

30

NoWait
Type

RECALL
Prod555

80

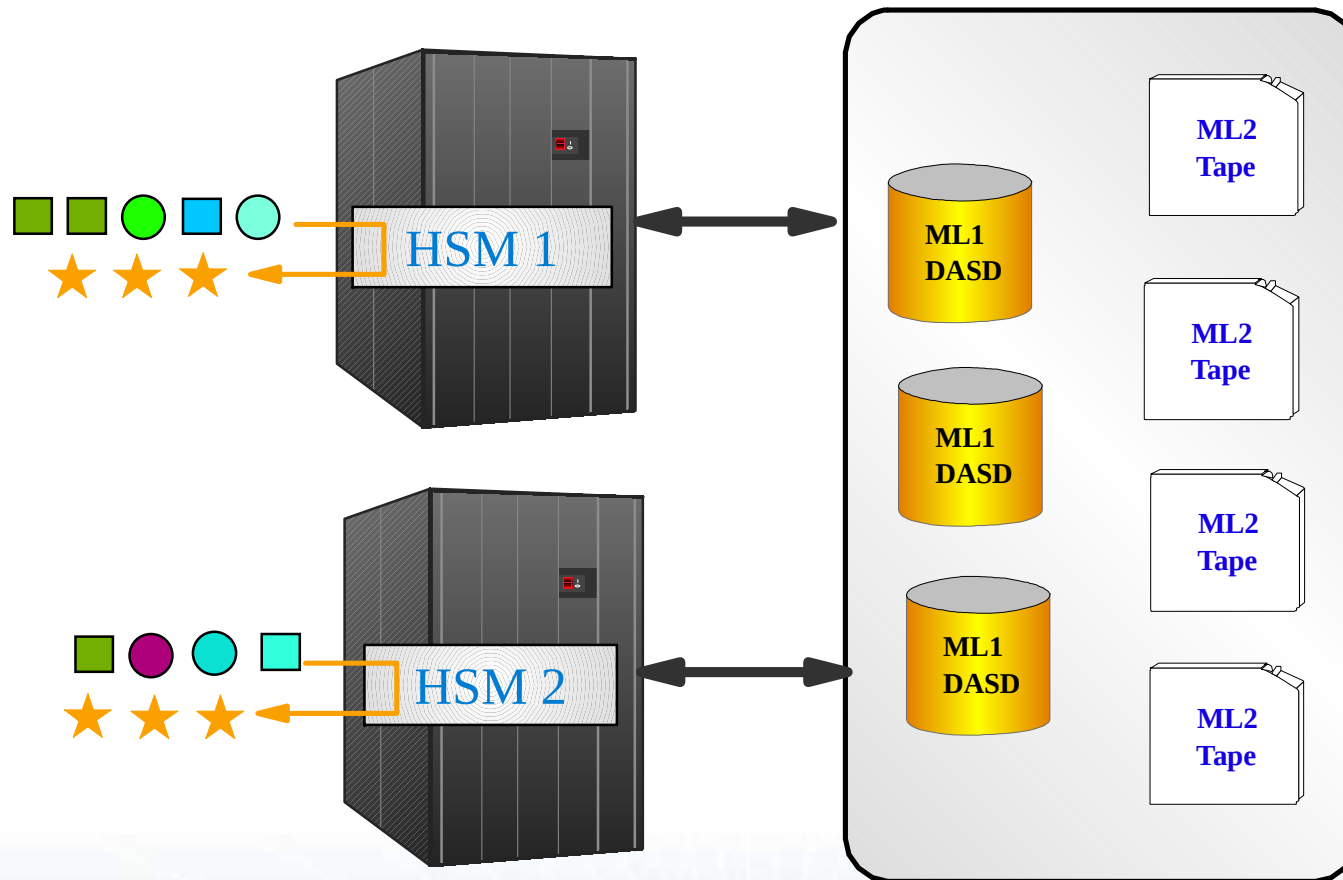
RECALL
Test678

30

DELETE
GDG234

10

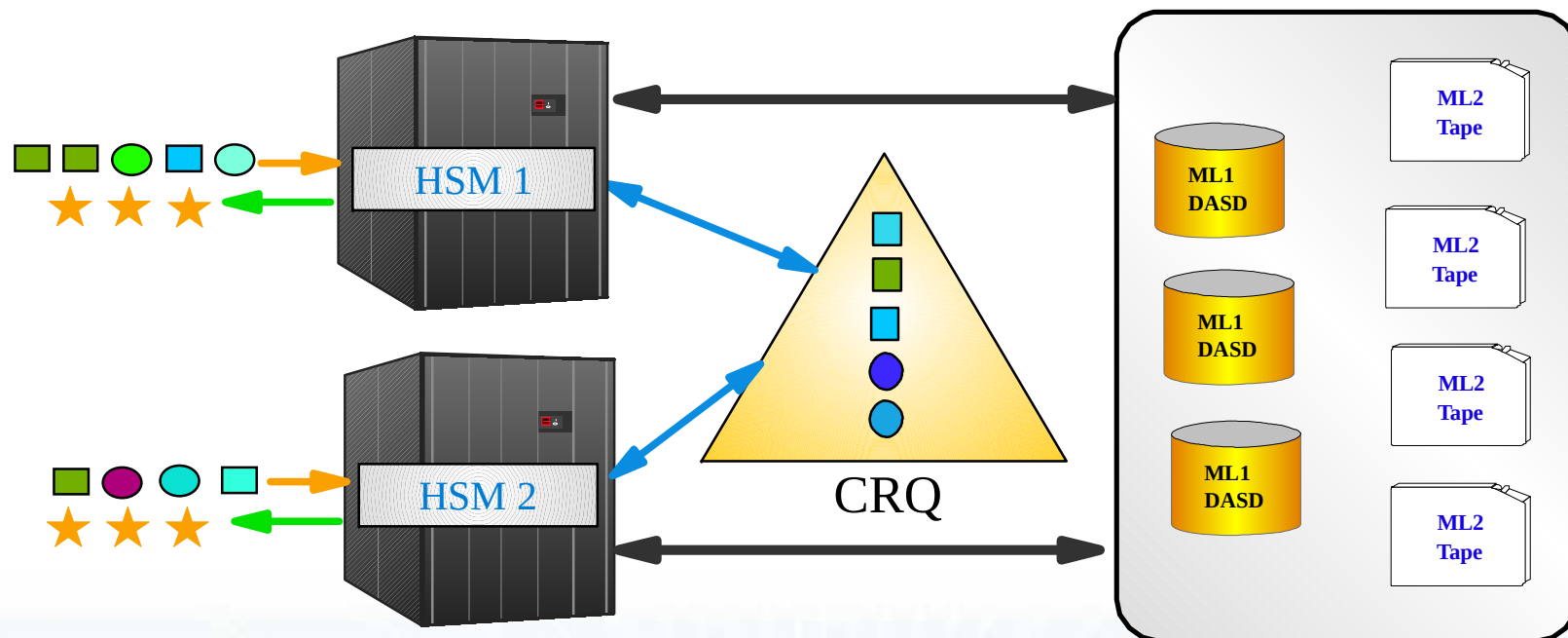
Recall in non-CRQ Environment



Common Recall Queue

All requests are placed onto a shared queue from which all hosts can process requests.

- Implemented using a CF List Structure.



CRQ Advantages



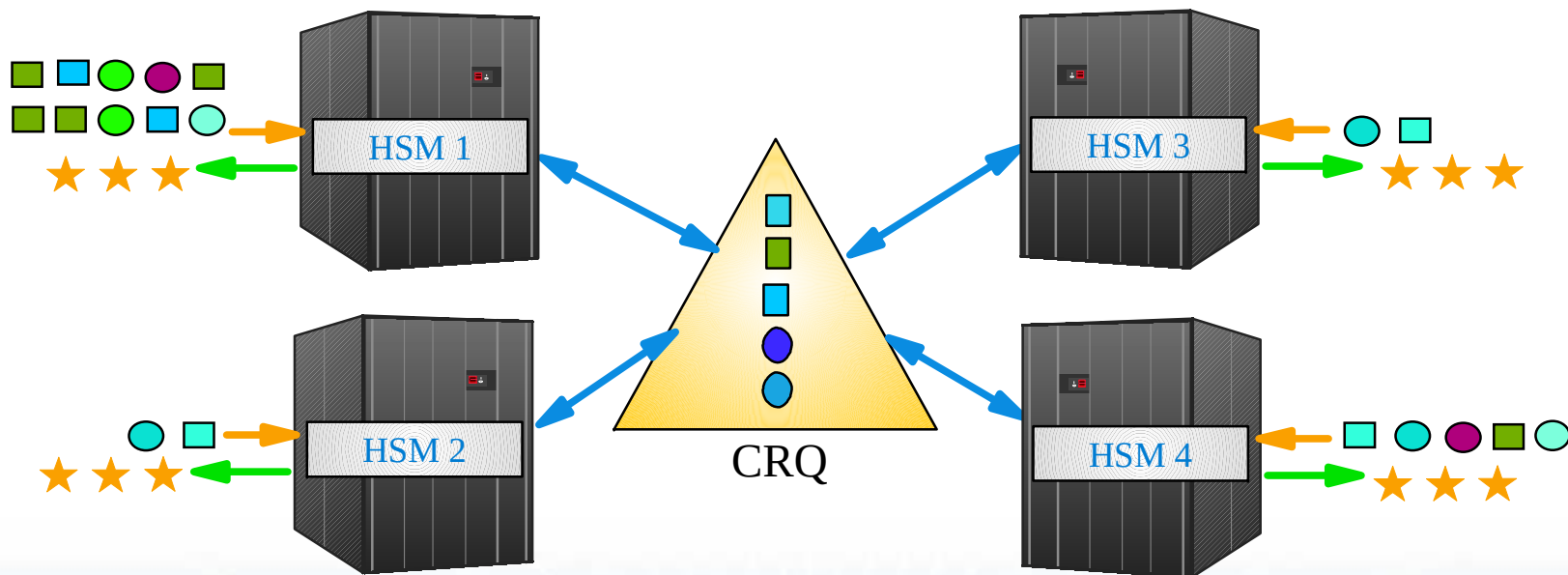
- Workload Balancing
- Tape-Mount Optimization
- Priority Optimization
- Flexible Configurations
- Request Persistence



Advantages

Work Load Balancing

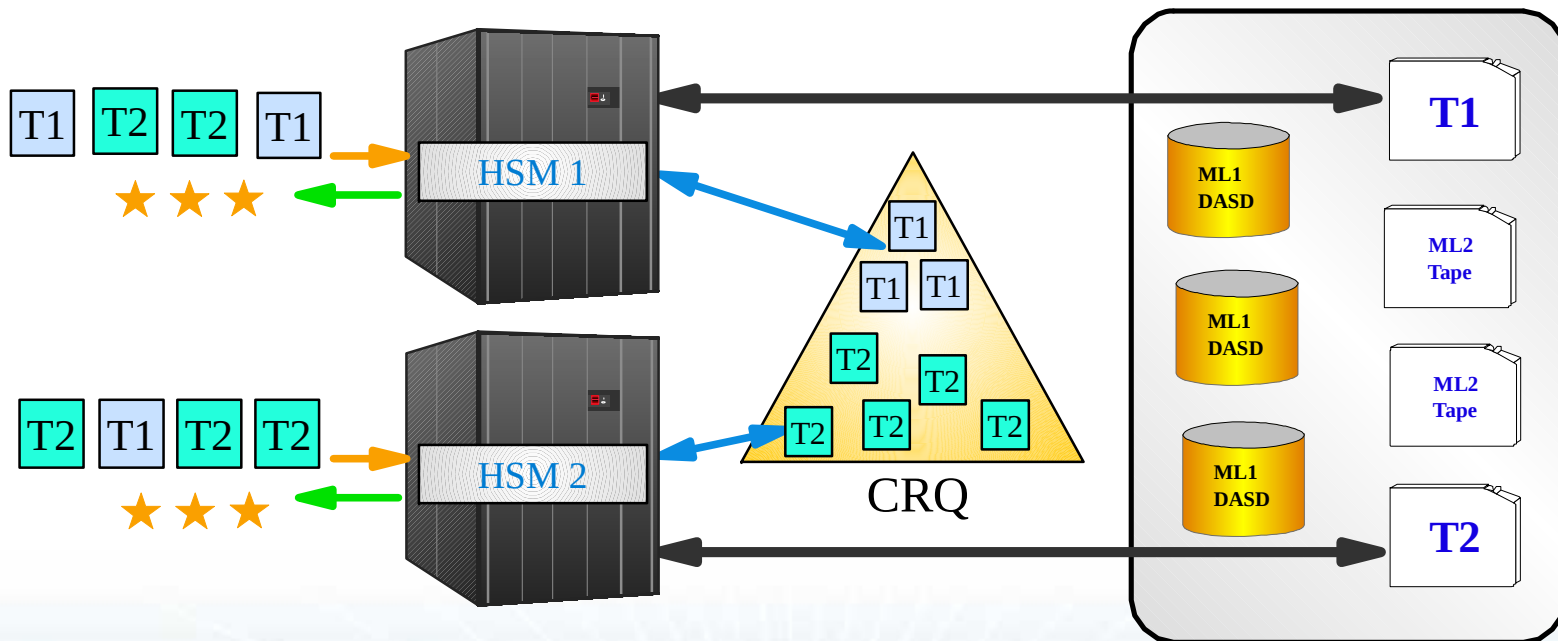
- ▶ All Recall requests are placed onto CRQ
- ▶ Requests are evenly distributed among hosts, up to the maximum number of Recall tasks



Advantages

Tape Mount Optimization

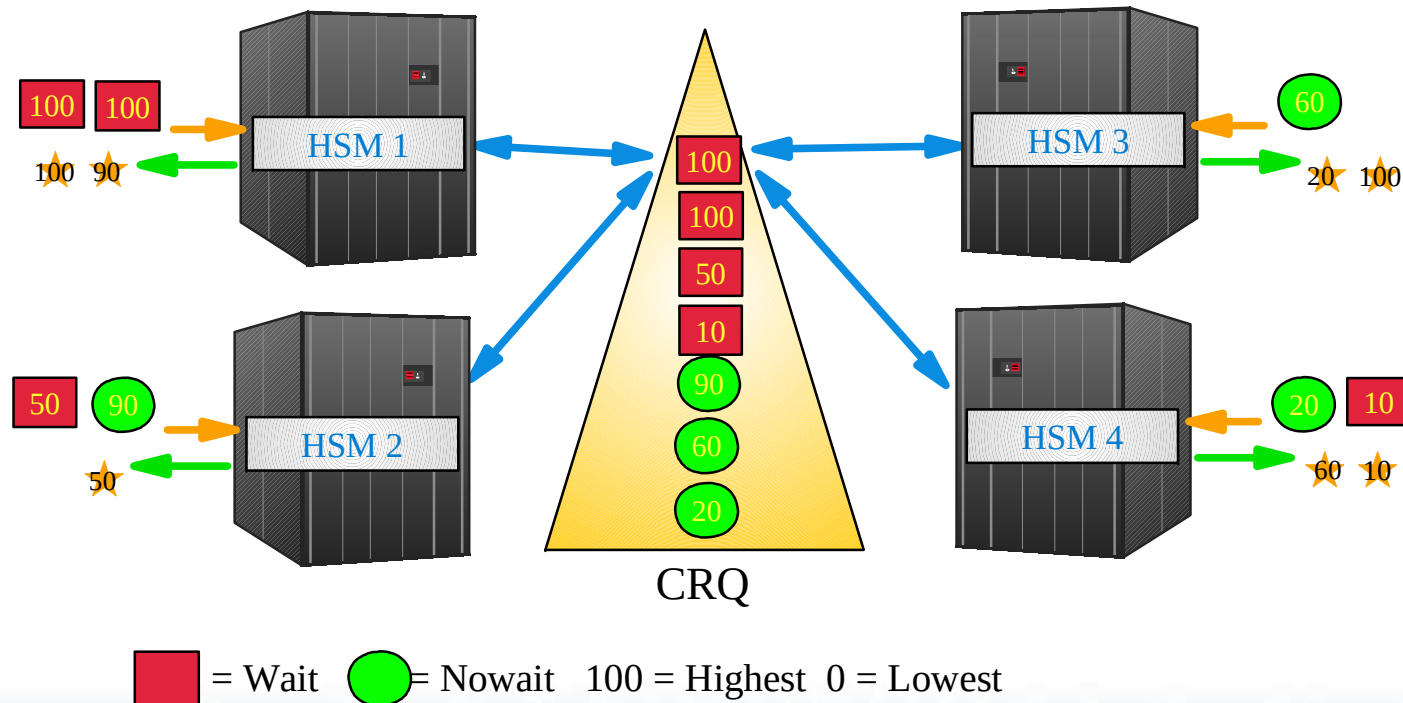
- ▶ A Recall task will process all requests in the CRQ that require the same tape
 - ★ Only a single tape mount is required



Advantages

Priority Optimization

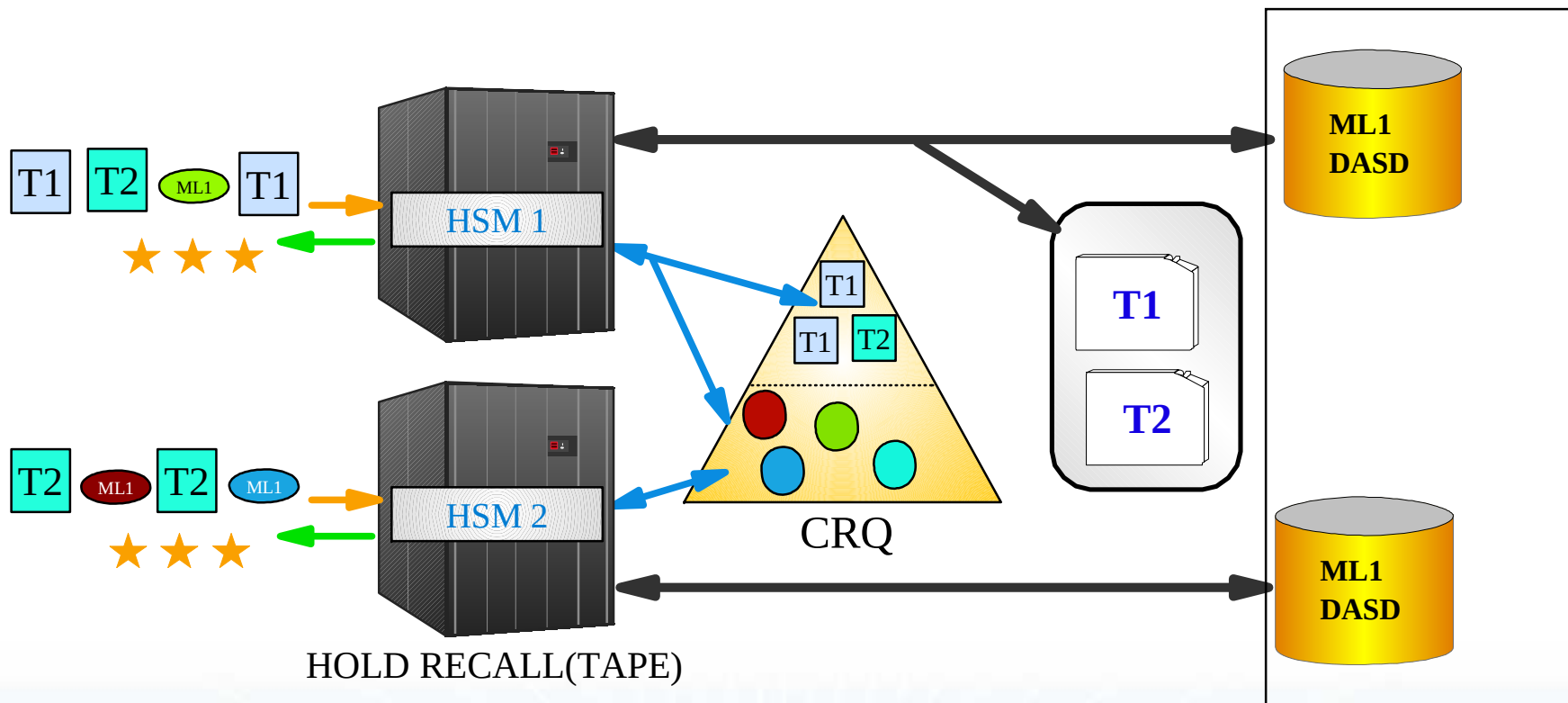
- ▶ Highest priority requests are always processed first



Advantages

Flexible Configurations - Tape

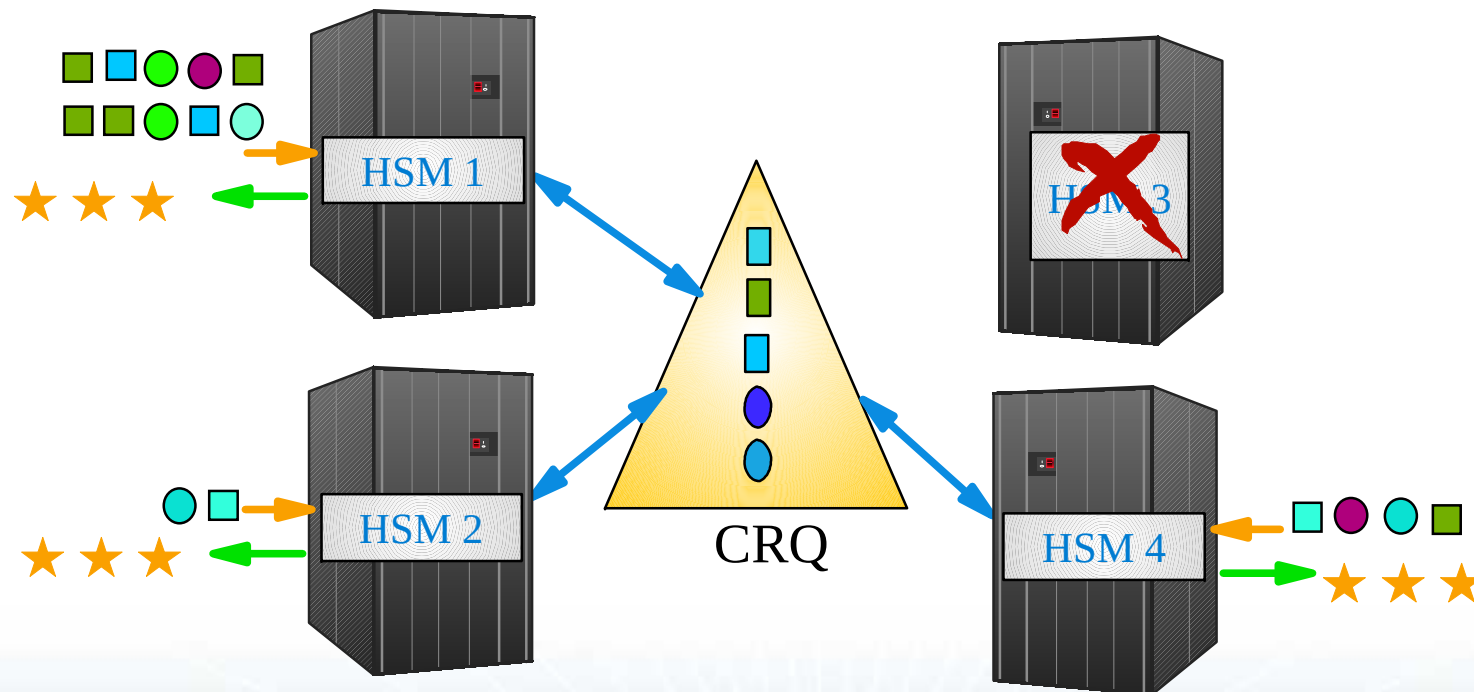
- ▶ Hosts not connected to tape drives can be configured to only select non-tape requests



Advantages

Flexible Configurations - Persistence

- ▶ Outstanding Recall requests from an unavailable host are processed by the remaining active hosts



Implementation



- Determine the size of the structure
- CFSIZER can be used
 - ▶ Interactive website for determining structure size
 - ▶ www.ibm.com/servers/eserver/zseries/cfsizer/
- Update the Coupling Facility Resource Manager policy
- Update DFSMSHsm parmlib
 - ▶ **SETSYS COMMONQUEUE(RECALL(CONNECT(name)))**
- No other changes required
- Functionally, use of CRQ is transparent to end-users

Error Handling in CRQ Environment



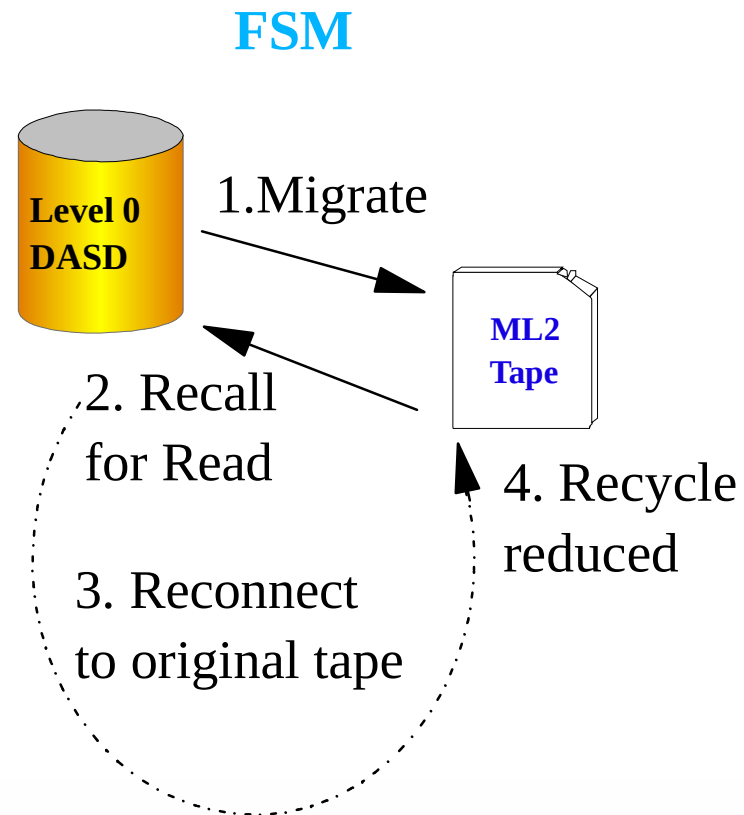
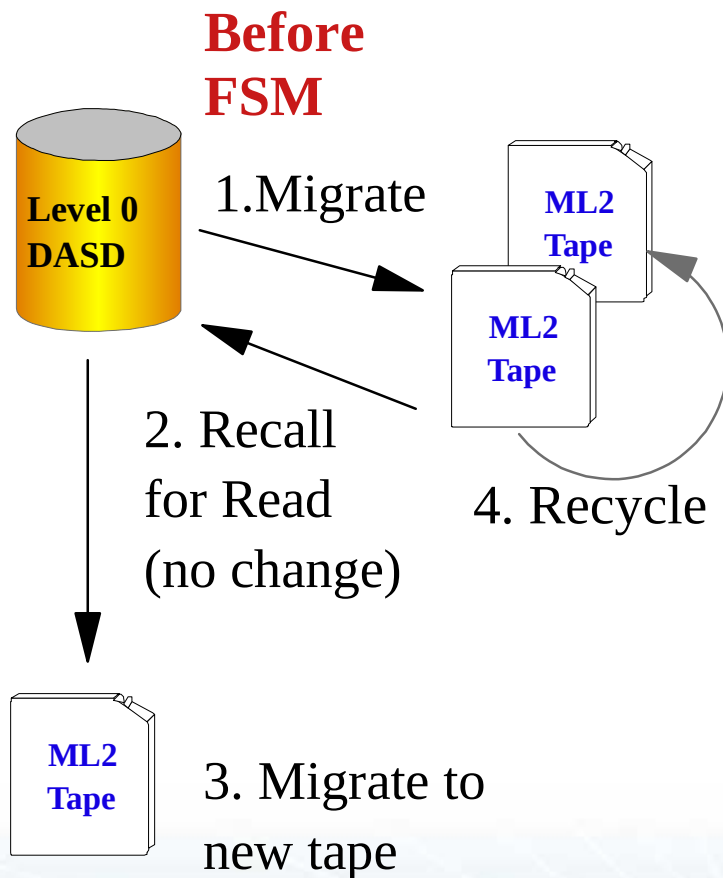
- If a z/OS image fails, the DFSMSHsm host on that system also fails, but all of the recall requests that originated on that host remain intact on the CRQ.
 - ▶ The coupling facility notifies the remaining connected hosts of the failure.
 - ▶ In-process requests on the failed host remain on the queue and are made available for other hosts to restart them.
- If the failing host was processing a request from ML2 tape, then recall requests for data on that tape cannot be selected
 - ▶ The tape is marked as "***in-use***" by the failing host
- The "in-use" indicator can be reset by:
 - ▶ Restarting the failed host or
 - ▶ Using the the **LIST HOSTID(hostid) RESET** command

Agenda



- Managing and Tuning the Recall Environment
- **Managing and Tuning the Migration Environment**
- Managing and Tuning the **HSM Control Data Sets** Environment
- Managing and Tuning the **HSM Tape** Environment
- Misc Tips and Hints

Fast Subsequent Migration



Fast Subsequent Migration



- New *SETSYS TAPEMIGRATION RECONNECT* keyword
 - ▶ RECONNECT(ALL)
 - ▶ RECONNECT(ML2DIRECTEDONLY)
 - ▶ RECONNECT(NONE)
- Reduces need to RECYCLE ML2 tapes
- Eligibility Determined at migration time
 - ▶ Primary Space Management (PSM)
 - ▶ Interval Migration (IM)
 - ▶ Data Set Migration
- ★ FSM was redesigned in DFSMSHsm V1R7 so it can be applicable to data sets that are not backed up

Agenda



- Managing and Tuning the Recall Environment
- Managing and Tuning the Migration Environment
- **Managing and Tuning the HSM Control Data Sets Environment**
- Managing and Tuning the HSM Tape Environment
- Misc Tips and Hints

CDS Performance Tuning



- Activate [CACHE](#) on MCDS, BCDS and OCDS
- Activate [DASD Fast Write](#) for Journal
 - ▶ Apply to 3990 Control Units
 - ▶ For ESS, both above are turned on by default and cannot be disabled
- Use [Record Level Sharing \(RLS\)](#) to allow CDSs to take advantage of cache structures in the Coupling Facility
 - ▶ CF region space = 360 KB per Host
 - ▶ Supports Extended Addressability >4GB
 - ★ Also Supported for non-RLS as of DFSMS V1R5
 - ▶ Supports Multi-cluster
 - ▶ Significant performance improvement for CDS I/O intensive activity such as SSM and EXPIREBV

Reorganizing your CDS's



- Reorg with FREESPACE(0 0) and let DFSMSHsm split midsection intervals
 - ▶ *Performance is degraded for about 2-3 weeks during this process*
 - ▶ *Do not panic when you see HURBA/HARBA ratio increase during first several days*
- ★ Only Reorg when you are increasing size allocation
- Make sure ALL hosts are shut down before attempting to reorg any CDS
- Use DISP=OLD on reorg job to prevent accidentally bringing up DFSMSHsm

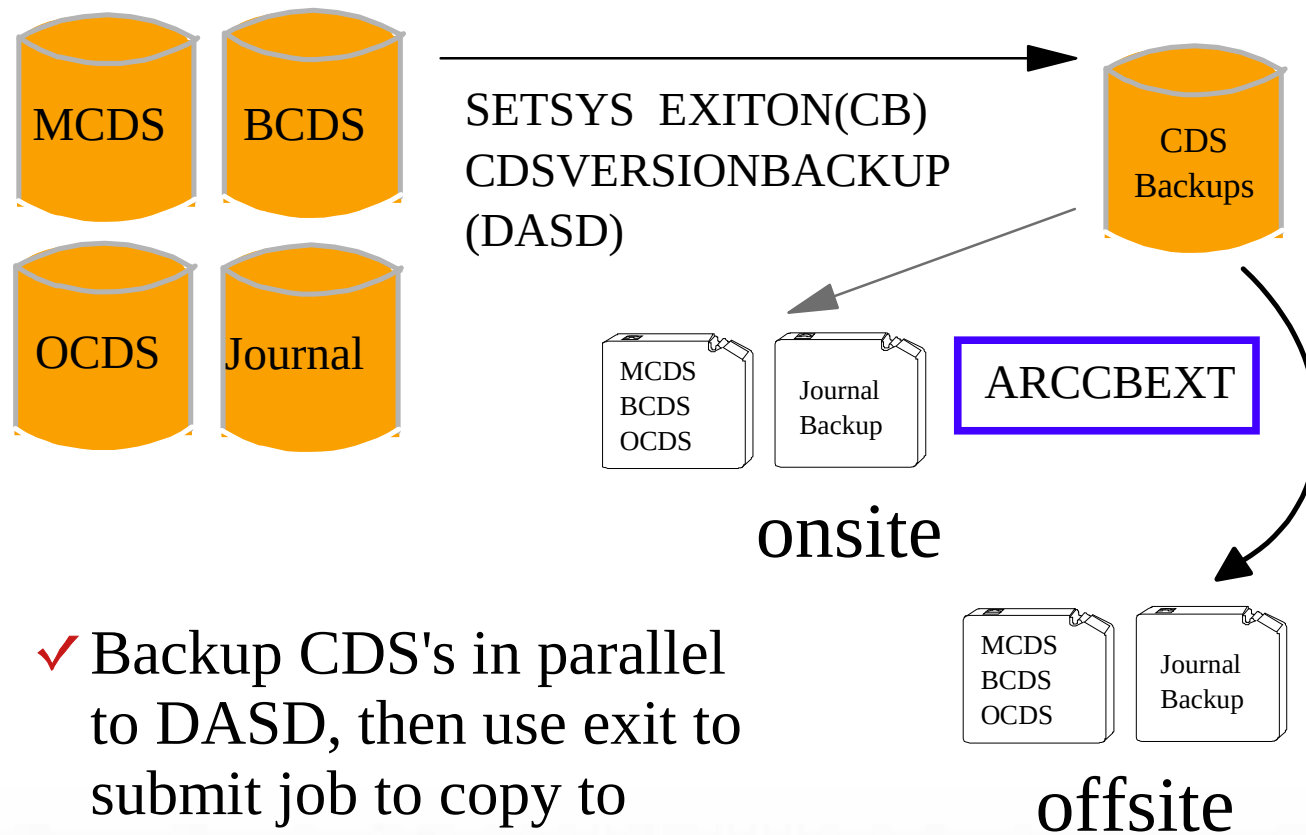
Multi-cluster MCDS and BCDS



- ✓ Keep each cluster on separate DASD subsystems to improve performance and reduce recovery time
 - ▶ Up to four (4) KSDS clusters can represent the MCDS
 - ▶ Up to four (4) KSDS clusters can represent the BCDS



Duplex your CDS Backups



- ✓ Backup CDS's in parallel to DASD, then use exit to submit job to copy to onsite and offsite tapes

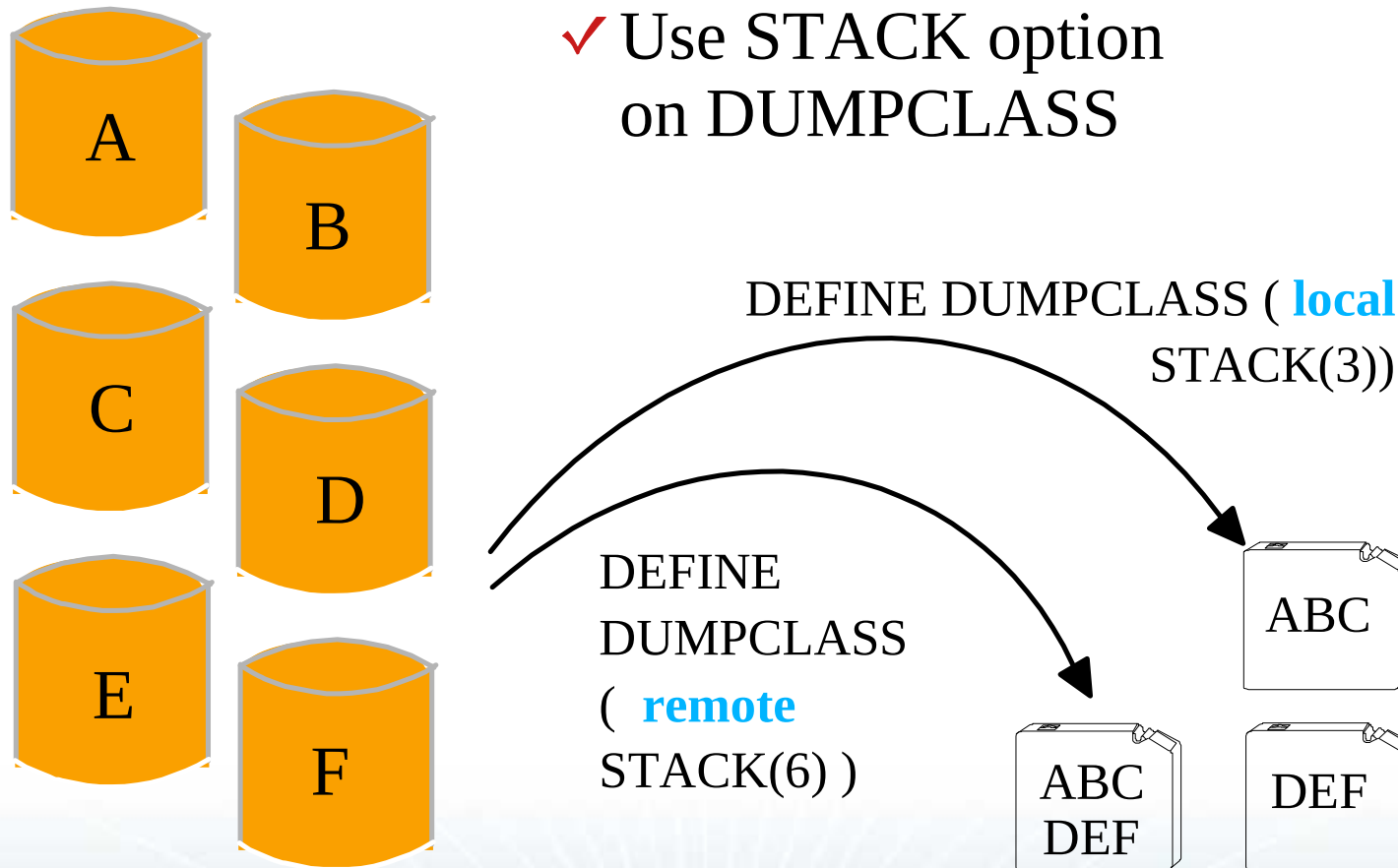
Agenda



- Managing and Tuning the Recall Environment
- Managing and Tuning the Migration Environment
- Managing and Tuning the HSM Control Data Sets Environment
- **Managing and Tuning the HSM Tape Environment**
- Misc Tips and Hints



Full Volume Dump Stacking

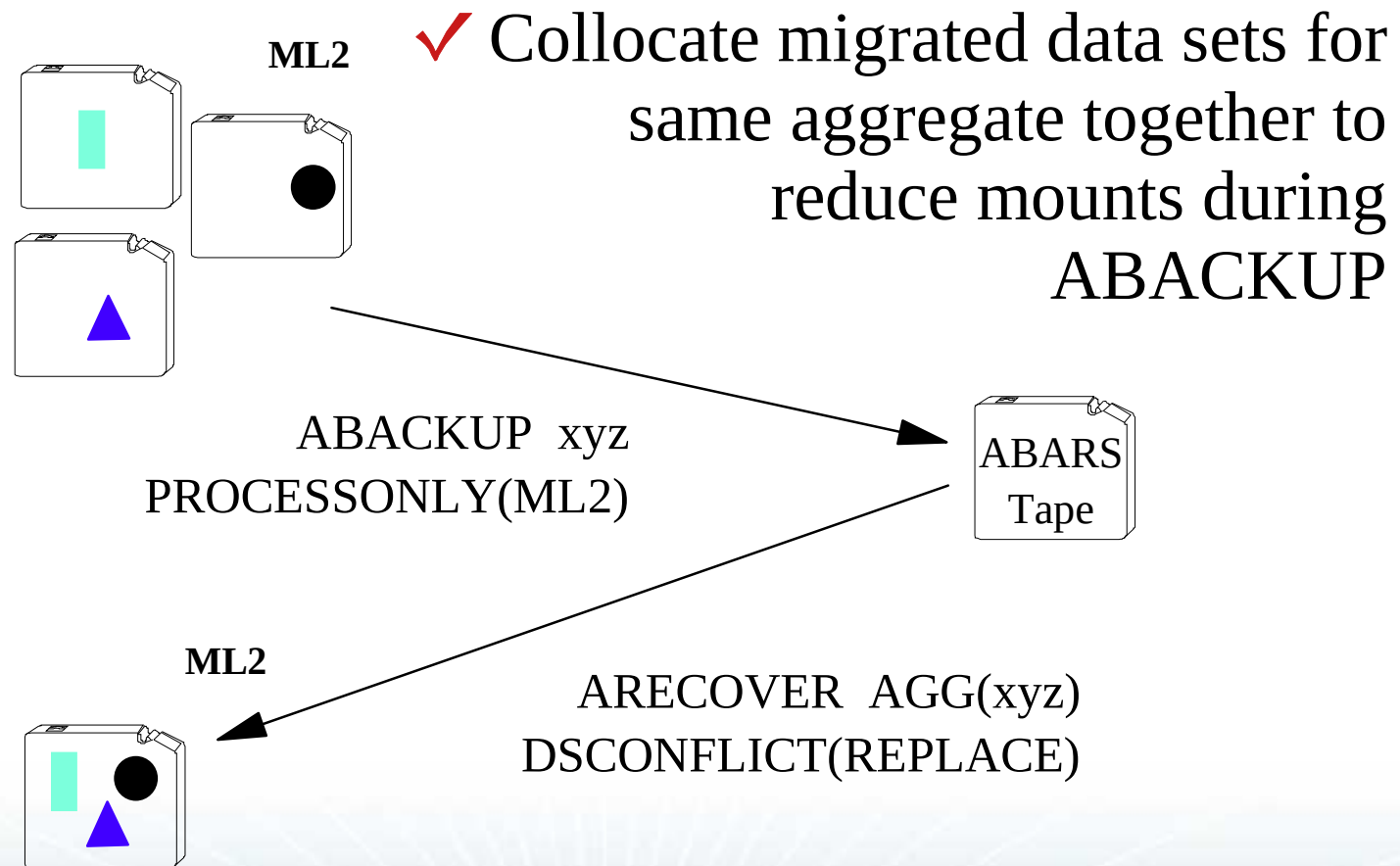


Recovering a volume from a stacked dump tape



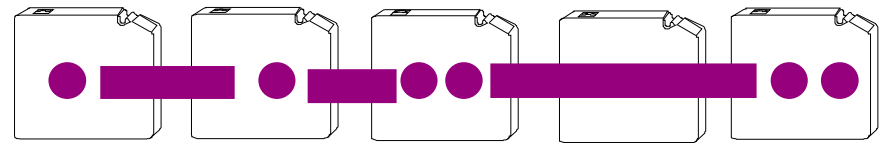
- How can I tell what dump volume DFSMShsm dumped a primary or ML1 volume to?
- How can I determine the file sequence number for that volume on a stacked dump volume?
 1. Issue a **LIST PVOL** or **LIST MVOL** with the **ALLDUMPS** parm to determine the dump volumes a primary volume or ML1 volume has been dumped to.
 2. Issue the **LIST DUMPVOLUME(volser) DUMPCONTENTS** command to determine which file sequence # on the dump tape that contains the dump of the particular migration or primary volume.
 3. Knowing this allows creation of JCL to restore the volume
 4. They can create a DFSMSdss batch job to restore the volume
 - ▶ Will require File Sequence Number in JCL
 - ▶ Can run multiple restores in parallel
 5. Issue the "**HSEND WAIT RECOVER * TOVOLUME(primary volser) UNIT(primary volser unittype) FROMDUMP(DUMPVOL(dump volser))**"
 - ▶ RECOVER volume with FROMDUMP are not multitasked

Collocate ML2 for ABARS



Avoid Connected Sets

- Connected Set - sequence of tape volumes connected by valid spanning data
 - ▶ Slows down recall and recycle activity
 - ▶ More difficult for tape library ejections
- You can minimize the occurrence of connected sets with the judicious use of **SETSYS TAPEUTILIZATION(PERCENTFULL)** and **TAPESPANSIZE** parameters
 - ▶ Never use TAPEUTILIZATION(NOLIMIT)
 - ▶ For TAPESPANSIZE see notes in discussion of new tape media
- You can break the connected set by doing one of the following to the spanning data set
 - ▶ Recall a migrated data set
 - ▶ Deleting a data set backup using the BDELETE command



Avoid Connected Sets



- You can determine if you have connected sets by issuing a **LIST TTOC SELECT(CONNECTED)**.
- Consider use of new **CHECKFIRST(N)** parameter on generic **RECYCLE** command if significant number of connected sets that meet PERCENTVALID criteria are not being recycled



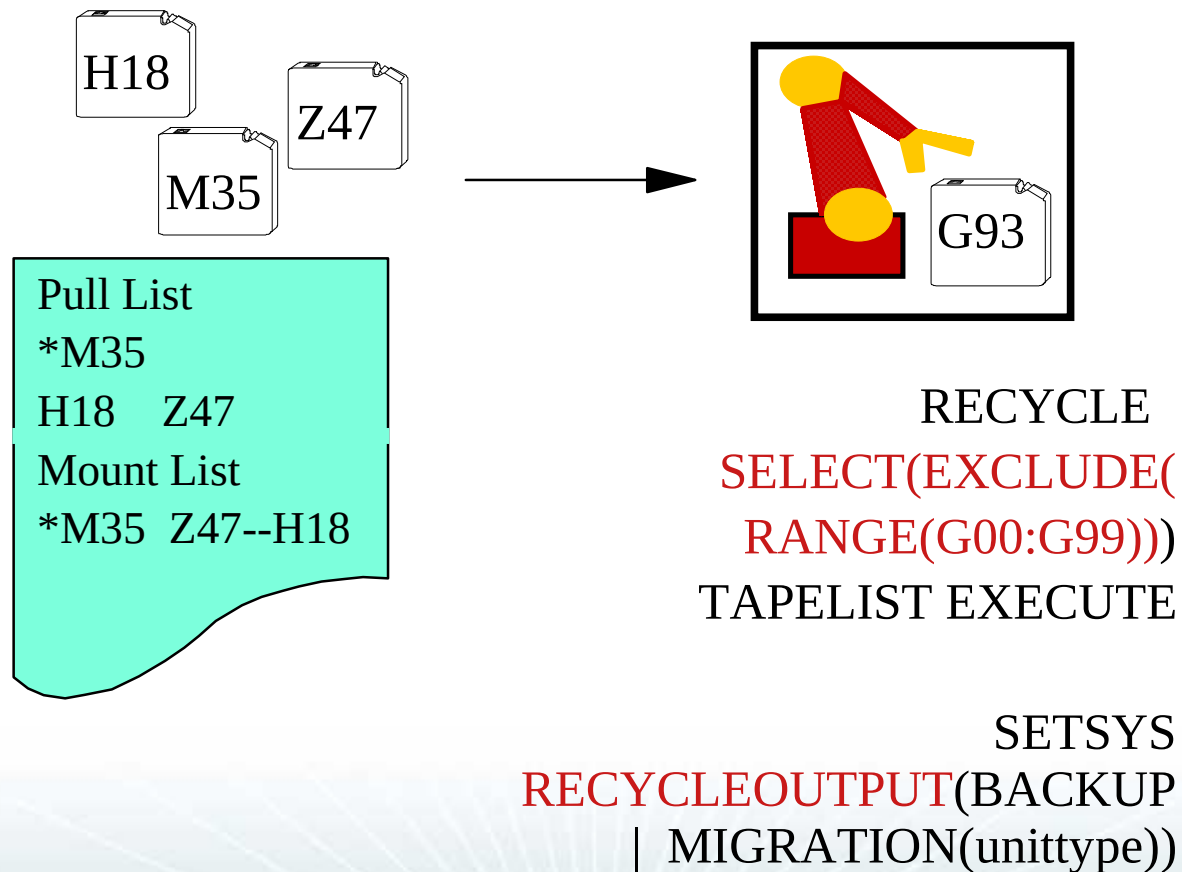
Tape and Recycle performance



- Use the **LIMIT** parameter to match **RECYCLE** workload to your scratch tape needs:
 - ▶ LIMIT(50) "Process enough input tapes to return a net gain of 50 scratch tapes"
 - ▶ Example: Read 60 input, create 10 output
- Use **PERCENTVALID(0)** to reclaim empty tapes when no drives available



Use Recycle to Convert to new tape technology



Use Recycle to create duplex tapes

- Issue **SETSYS DUPLEX(MIGRATION(Y))** and/or **SETSYS DUPLEX(BACKUP(Y))** to begin duplexing migration and backup tapes
- You may want to specify **PERCENTVALID** to a high value on the generic recycle command if you want the duplexing to take effect more quickly!
- If **PERCENTVALID** not specified on the generic **RECYCLE** command, then:
 - ▶ When processing ML2 volumes, DFSMSHsm uses the **ML2RECYCLEPERCENT** value that has been specified on the **SETSYS** command.
 - If **ML2RECYCLEPERCENT** has not been specified, DFSMSHsm uses the **SETSYS RECYCLEPERCENT** value.
 - If **RECYCLEPERCENT** has not been specified, the DFSMSHsm default is 20%.
- **TAPECOPY** is another method for creating duplex tapes but has the following limitations
 - ▶ Single threaded task structure
 - ▶ Is interruptable by recall, but if interrupted, must be restarted from beginning

Speeding up RECYCLE when duplexing



- APAR: OW45264
- Bypassing the SYNCDEV for the duplex alternate tape during Recycle is now supported.
 - ▶ This reduces the time Recycle takes by bypassing the SYNCDEV of the duplex alternate tape after each dataset.
 - ▶ To bypass the alternate tapes SYNCDEV operation, a new HSM PATCH may be issued.

PATCH .YGCB.+C8 BITS(.....1.)

- ▶ This patch is not recommended for earlier technology tape drives such as the 3490 and 3590.

Tape Device Overview



■ Recent Media Types - *More to Come*

- ▶ Media 5 - up to 500 GB R/W
- ▶ Media 6 - up to 500 GB WORM
- ▶ Media 7 - up to 100 GB R/W
- ▶ Media 8 - up to 100 GB WORM

★ *All capacities are physical capacity without compaction, EFMT2 recording technology on 3592 E02 drives*



Tape Device Overview



- Newer drives allow more data on each tape
 - ▶ 3592-E05 = 100GB (Media7) or 500GB (Media5) (physical)
 - ▶ 3592-J1A = 60GB (Media7) or 300GB (Media5) (physical)
 - ▶ 3590-H1x = 30GB (Media3) or 60GB (Media4) (physical)

- Recording technology
 - ▶ 3592-E05 = EFMT2
 - ▶ 3592-J1A = EFMT1
 - ▶ 3590-H1x = 384 tracks



Device Overview (*continued*)



- 3592 E05 devices must be genned as device type 3590.
 - ▶ They can read and write in both EFMT1 and EFMT2.
- 3592 E05 devices can write
 - ▶ to full tape capacity (MEDIA5,6,7,8),
 - ▶ to 20% capacity with **performance scaling** (MEDIA5)
 - ▶ write a tape in two segments with **performance segmentation** (MEDIA5).

Making 3592 E05's Known to HSM



- Install enabling APAR OA11388 (V1R4, V1R5) or APAR OA13110 (V1R6, V1R7).
- Define esoterics in the system IODF if in a non-SMS mixed environment.
- Define esoteric to HSM via SETSYS UUT(esoteric1:esoteric1...)
- Tell HSM to use the unit name i.e. SETSYS BACKUP(TAPE(esoteric1))



Activating Performance Segmentation or Scaling



- A data class must exist with the **Performance Segmentation** or **Performance Scaling** attribute set to 'Y'.
 - ▶ The ACS routine must map a DFSMShsm single file tape data set name to the data class.
 - ▶ Since data class now determines if a Performance option is used, non-SMS tape needs ACS routines if a 3592 Performance option is desired.
- IEC205I at Close of tape will indicate if Performance Segmentation or Scaling was used.
- Performance Segmentation and Performance Scaling only apply to **Media5**
- Performance Segmentation and Performance Scaling are mutually exclusive.

Where to use Performance Scaling



- Where read response time of individual data sets is critical such as Recall of migrated data sets.
 - ▶ Besides faster seek times, this also allows more recall tasks to operate simultaneously across multiple tape drives.
- Where recovery time of many data sets is critical such as disaster recovery.
 - ▶ Splitting recovery across ten smaller tapes running on ten tape drives is much faster than recovering two larger tapes on two tape drives.



Where to use Performance Segmentation



- Where a compromise between higher performance and higher capacity is desired.
- First 20% of tape has faster access time than a non-segmented tape.
- Write performance is improved in the last few wraps of the tape since RABF scratch region is not overwritten by user data.



Exploitation of Performance cartridges



- DFSMShsm can support a *tape hierarchy strategy*
 - ▶ You can use the shorter length 3592 tapes (or a performance scaled 300 GB tape (scaled to 60 GB)) for the initial ML2 tapes
 - ▶ When recycling, write to the longer length (or full capacity) 3592 tapes.
- To do this (since the data set names would be the same), you would use the
 - ▶ **SETSYS RECYCLEOUTPUT (MIGRATION(unittype))**
 - ▶ Specify an esoteric that in your ACS filter routines selects a data class that would then use the longer length tapes (or tapes that aren't performance scaled)
- When migrating to ML2, you would need to select a dataclass that specifies the shorter length tape or uses the performance scaling option with MEDIA5.

Error Handling for Duplex Alternate



- **APAR OA09279** - Error Handling for Duplex Alternate
 - ▶ Will continue to discard alternate tape in error
 - ▶ Tapecopy will continue to be scheduled
 - ▶ Original Tape will be demounted at start of next dataset
 - Allows internally scheduled TAPECOPY to occur sooner
 - Less data will need to be copied
 - ▶ Duplexing will be resumed with new original/alternate combination
 - ▶ New message ARC0953I issued when failed alternate discarded
 - Allows installation to mark tape for permanent retention
 - ▶ Installations can later issue FIXCDS to original tape's TTOC if access to the alternate is required
 - ▶ Designed to reduce disparity between original/alternate combination is the one data set in process at the time of failure
 - ▶ Once TAPECOPY completed, both tapes should be identical
 - ▶ The following DFSMSHsm patch command will activate the changes in this APAR:

PATCH .MCVT.+594 BITS (.....1.)

DFSMSHsm and VTS



- In a VTS environment

- ▶ **SETSYS PARTIALTAPE** should be **MARKFULL** and not **REUSE**.
 - **REUSE** causes the complete virtual volume to be staged when DFSMSHsm goes to extend it.
 - Perhaps worse yet, creates a hole in the physical tape from which the virtual tape came.
 - Only used portion of virtual volume de-staged to back-end tape.
 - **MARKFULL** however, does increase number of virtual volumes required for DFSMSHsm.

DFSMSHsm and VTS

- If using the new **4GB virtual volume** in a VTS
 - ▶ You have to specify a multiplier value to inform DFSMSHsm that the capacity of the logical 3490 is greater than 400MB or 800MB.
 - **SETSYS TAPEUTILIZATION(LIBRARYBACKUP)** for backup
 - **SETSYS TAPEUTILIZATION(LIBRARYMIGRATION)** for migration
 - ▶ The multiplier will vary depending on whether the Data Class defines the DFSMSHsm backup tapes as CCST (400MB) or ECCST (800MB) capacity.

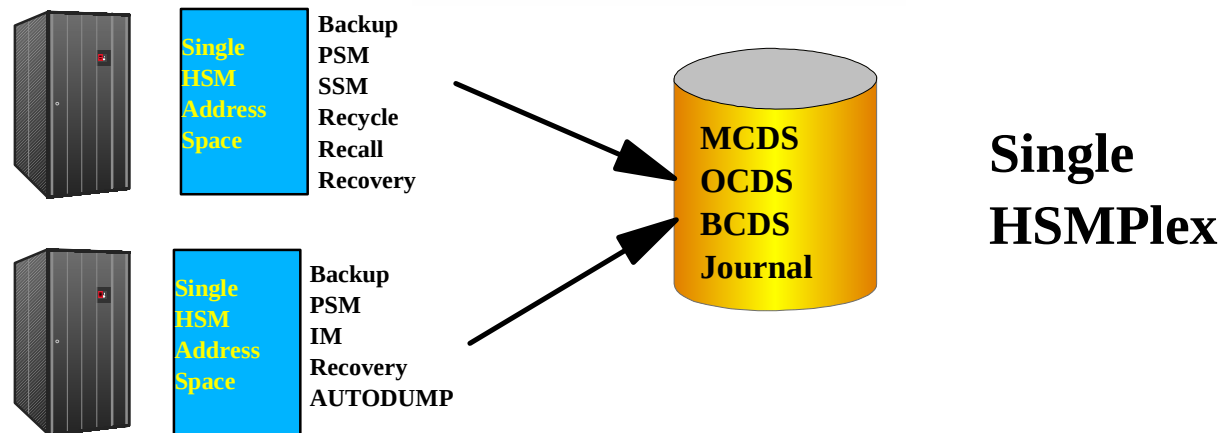
Agenda



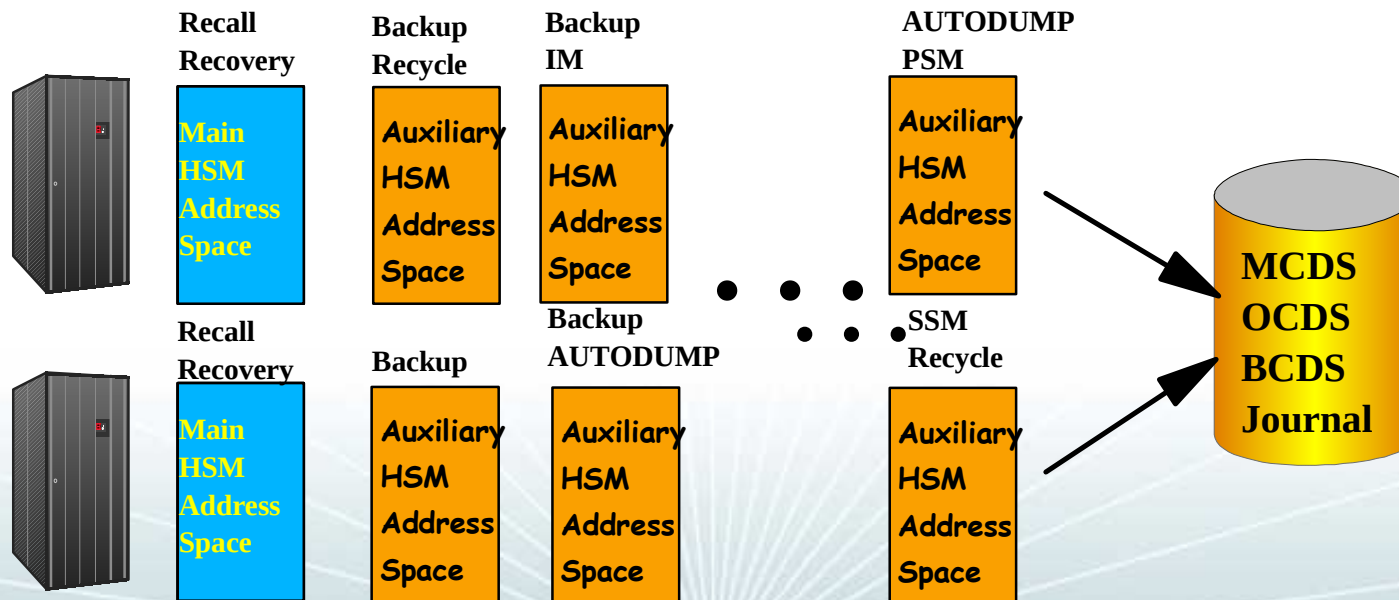
- Managing and Tuning the Recall Environment
- Managing and Tuning the Migration Environment
- Managing and Tuning the Backup Environment
- Managing and Tuning the HSM Control Data Sets Environment
- Managing and Tuning the HSM Tape Environment
- **Misc Tips and Hints**

Multiple Address Space HSM

non-MASH



MASH



Multiple Address Space



■ Multiple Address Space Support

- ▶ Each Address Space runs as unique DFSMSHsm host
- ▶ Up to 39 DFSMSHsm hosts per HSMplex

■ Potential Benefits

- ▶ Different hosts can be assigned different functions
- ▶ Different hosts can be assigned different Workload Manager (WLM) Velocity Goals
- ▶ Reduces SYSZTIOT contention for DASD/Tape allocations
- ▶ Improved availability
- ▶ Increased Tasking Levels of most functions in DFSMSHsm



MAIN versus AUX hosts



■ HOSTMODE=MAIN

- ▶ One per system image
- ▶ Handles implicit recalls, HSEND commands, batch and TSO end user requests
- ▶ Manages ABARS secondary address spaces
- ▶ This is the default HOSTMODE

■ HOSTMODE=AUX

- ▶ Multiple per system image
- ▶ Scheduled Automatic Functions
 - PSM, SSM, Interval
 - Autobackup
 - AUTODUMP
- ▶ Explicit requests using Console Commands, Netview or EMCS

CRQ and MASH



- CRQ allows AUX host to also handle implicit recall requests
- You can define "Recall Servers" in CRQ by
 - ▶ Issuing **HOLD COMMONQUEUE(RECALL(SELECTION))** on hosts that you do not want to select recall requests off the CRQ
- You can use WORKLOAD Manager to assign higher velocity goals to MASH host(s) that are recall servers



SMF consolidation processing



- ✓ Specify **DDCONS(NO)** on SMF parameters to avoid DD name consolidation during shutdown
 - ▶ DDCONS is specified in SMFPRMnn parmlib member
 - ▶ See MVS Initialization and Tuning Reference for more information
- DFSMShsm is a started task with thousands of DD name allocations
- DFSMShsm shutdown may be delayed up to 45 minutes if consolidation is performed
- SMF Type 30 records are a bit longer

Avoid LOG overhead



- ✓ Use **HOLD LOG** to avoid DFSMSHsm logging overhead
 - Command can be added to PARMLIB
 - Turns off writing to the LOGX/LOGY data sets
 - Information available elsewhere, such as FSR records in SMF, Activity Logs, PDA trace data
 - Some ISV products require the LOGX/LOGY data sets as input
 - Reduces DFSMSHsm overhead activity (approximately 7% reduction)

Use of ONLYIF in parmlib member



► ONLYIF HSMHOST(hostid)

- Conditional processing based on Host identifier
- Specify only the 1-character hostid, not the Y or N primary-host identifier
- Single command following ONLYIF executed based on match on **HOST=xy** parameter in startup procedure
- Useful for sharing ARCCMDnn across systems

► Example

```
ONLYIF HSMHOST(1)
```

```
    SETSYS PRIMARYSPMGMTSTART(400 730)
```

```
ONLYIF HSMHOST(2)
```

```
    SETSYS PRIMARYSPMGMTSTART(900 1100)
```

Misc hints:



- Put **PDA=YES** in DFSMSHsm Proclib
 - This allows tracing to start earlier during DFSMSHsm startup
- Pay attention to "Level1 Days Non-Usage" and "Primary Days Non-Usage" attributes in Management Class
 - Level1 Days Non-Usage value includes time spent on Primary
 - For example
 - Primary Days Non-Usage = 4
 - Level1 Days Non-Usage = 3
 - Results in data set being migrated to ML2 after 4 days and never being migrated to ML1
 - In this case you probably want Level1 Days Non-Usage = 7

Parmlib recommendation



- ▶ SPECIFY EVERYTHING!!!
- ▶ Defaults change with maintenance and new releases
- ▶ Improved change control if all commands appear in ARCCMDnn
- ▶ One-stop setup verification



DFSMSHsm Journal and EDI



- z/OS V1R5 introduced an enhancement in called "**Enhanced Data Integrity**" or EDI.
 - ▶ Provides for additional integrity checking for PS (physical sequential) data sets.
 - ▶ Activated by the customer creating a new SYS1.PARMLIB member, IFGPSEDI, and specifying either MODE(WARN) (issue warning message) or MODE(ENFORCE) (abend).
 - ▶ If IFGPSEDI doesn't exist or MODE(WARN) or MODE(ENFORCE) isn't specified, EDI will not be active and current processing for DISP=SHR PS data sets will continue.

DFSMSHsm Journal and EDI



- ▶ If EDI is active, those applications that can maintain data integrity on their own, when their data sets are accessed for output concurrently by multiple users, will be allowed to specify those data set names in new PARMLIB member - IFGPSEDI
- ▶ The *DFSMSHsm journal* is a shared (for update) sequential data set. If the user activates EDI and they take no other action, *DFSMSHsm startup will fail*.
- ▶ If the customer has activated EDI, they need to place the name(s) of their DFSMSHsm journal data set in PARMLIB member IFGPSEDI

Summary



- Work Smarter
- Improve Performance
- Reduce Contention
- Simplify Handling
- Exploit New Functions
- Exploit Technology